

Finding the Domain of a Function

Example with a Quadratic Expression in the Denominator

Written in British English, with simple step-by-step feedback

Example chosen

The function is:

$$f(x) = \frac{x^2 - 3}{\sqrt{x^2 - 8x + 15}}$$

We need to find the **domain** of the function.

Aim

We need to decide which values of x make the function defined.

In this function, there are two important points to consider. First, there is a **denominator**, so the denominator cannot be equal to zero. Secondly, there is a **square root**, so the expression inside the square root must be greater than or equal to zero.

However, because the square root is in the **denominator**, the expression inside the square root must actually be **strictly greater than zero**.

Step-by-step method

Step 1: Apply the denominator condition

The denominator is:

$$\sqrt{x^2 - 8x + 15}$$

Since division by zero is undefined, we must have:

$$\sqrt{x^2 - 8x + 15} \neq 0$$

Feedback if the student makes a mistake:

Remember the two main rules. The denominator cannot be equal to zero. Before solving anything, write down the condition that prevents division by zero.

Step 2: Apply the square root condition

Since there is a square root, the expression inside the square root must be greater than or equal to zero:

$$x^2 - 8x + 15 \geq 0$$

Feedback if the student makes a mistake:

Remember that the expression inside a square root must be greater than or equal to zero. First, identify what is inside the square root, then write the correct inequality.

Step 3: Combine the denominator and square root conditions

We have:

$$\sqrt{x^2 - 8x + 15} \neq 0$$

and:

$$x^2 - 8x + 15 \geq 0$$

The square root is different from zero exactly when the expression inside the square root is different from zero:

$$x^2 - 8x + 15 \neq 0$$

So we need:

$$x^2 - 8x + 15 \geq 0$$

and:

$$x^2 - 8x + 15 \neq 0$$

Therefore:

$$x^2 - 8x + 15 > 0$$

Feedback if the student makes a mistake:

Be careful: when you combine the two conditions, both must be satisfied at the same time. Check that your final condition respects both the square root rule and the denominator rule.

Step 4: Solve the associated quadratic equation

To solve:

$$x^2 - 8x + 15 > 0$$

we first solve the associated equation:

$$x^2 - 8x + 15 = 0$$

Feedback if the student makes a mistake:

To solve a quadratic inequality, first solve the related quadratic equation. This gives you the key values to place on the number line.

Method 1: Using the quadratic formula

The quadratic formula is:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

In this case:

$$a = 1, \quad b = -8, \quad c = 15$$

Substitute these values into the formula:

$$x = \frac{-(-8) \pm \sqrt{(-8)^2 - 4(1)(15)}}{2(1)}$$

$$x = \frac{8 \pm \sqrt{64 - 60}}{2}$$

$$x = \frac{8 \pm \sqrt{4}}{2}$$

$$x = \frac{8 \pm 2}{2}$$

Now find the two solutions:

$$x = \frac{8 - 2}{2} = \frac{6}{2} = 3$$

$$x = \frac{8 + 2}{2} = \frac{10}{2} = 5$$

So the two solutions are:

$$x = 3 \quad \text{or} \quad x = 5$$

Feedback if the student makes a mistake:

Check the values of a , b , and c carefully. Pay special attention to the sign of b before substituting into the quadratic formula.

Method 2: Using factorisation

In this case, we can also factorise:

$$x^2 - 8x + 15$$

We need two numbers that multiply to 15 and add to -8 .

These numbers are -3 and -5 , so:

$$x^2 - 8x + 15 = (x - 3)(x - 5)$$

Now solve:

$$(x - 3)(x - 5) > 0$$

The critical values are:

$$x = 3 \quad \text{and} \quad x = 5$$

This method is quicker than the quadratic formula, but it is only straightforward when the quadratic factorises easily.

Feedback if the student makes a mistake:

Look for two numbers that multiply to the constant term and add to the coefficient of x . Be careful with negative signs.

Step 6: Decide where the quadratic is positive

The critical values are:

$$3 \quad \text{and} \quad 5$$

These split the number line into three intervals:

$$x < 3$$

$$3 < x < 5$$

$$x > 5$$

Since the coefficient of x^2 is positive, the parabola opens upwards. Therefore, the expression is positive outside the two roots and negative between them.

So:

$$x^2 - 8x + 15 > 0$$

when:

$$x < 3 \quad \text{or} \quad x > 5$$

Feedback if the student makes a mistake:

The values you found split the number line into intervals. Check which intervals make the quadratic positive. Remember, the question is asking for values greater than zero.

Alternative Step 6: Using a sign table

Using:

$$(x - 3)(x - 5) > 0$$

The critical values are:

$$3 \quad \text{and} \quad 5$$

For each interval, check the sign of each bracket and then combine the signs.

For $x < 3$:

$$(x - 3) < 0 \quad \text{and} \quad (x - 5) < 0$$

Therefore:

$$(-)(-) = +$$

So the expression is positive.

For $3 < x < 5$:

$$(x - 3) > 0 \quad \text{and} \quad (x - 5) < 0$$

Therefore:

$$(+)(-) = -$$

So the expression is negative.

For $x > 5$:

$$(x - 3) > 0 \quad \text{and} \quad (x - 5) > 0$$

Therefore:

$$(+)(+) = +$$

So the expression is positive.

Therefore, the solution is:

$$x < 3 \quad \text{or} \quad x > 5$$

Feedback if the student makes a mistake:

Choose one test value from each interval. Check whether each bracket is positive or negative, then combine the signs.

Step 7: Write the final domain

The domain is:

$$x \in \mathbb{R} \text{ such that } x < 3 \text{ or } x > 5$$

In interval notation:

$$(-\infty, 3) \cup (5, +\infty)$$

Feedback if the student makes a mistake:

Check whether the end points should be included or excluded. Since the denominator cannot be zero, any value that makes the denominator zero must be excluded.

Important note for AI feedback

A useful AI feedback system should not immediately give the answer. It should first identify which type of error the student has made.

If the student forgets the denominator condition, the feedback should remind them to check whether any part of the function is in the denominator.

If the student writes $x^2 - 8x + 15 \geq 0$ as the final condition, the feedback should prompt them to think about whether the square root is allowed to be zero when it is in the denominator.

If the student only finds $x = 3$ and $x = 5$, the feedback should explain that these are critical values, not the final solution of the inequality.

If the student includes 3 or 5 in the final domain, the feedback should remind them that the denominator cannot be zero, so those values must be excluded.

If the student chooses the interval $3 < x < 5$, the feedback should suggest checking the sign of the quadratic on each interval or thinking about the direction of the parabola.